

Performance Report for main-ec-16K-soko07-2+2-3osd

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Summary of results for main-ec-16K-soko07-2+2-3osd

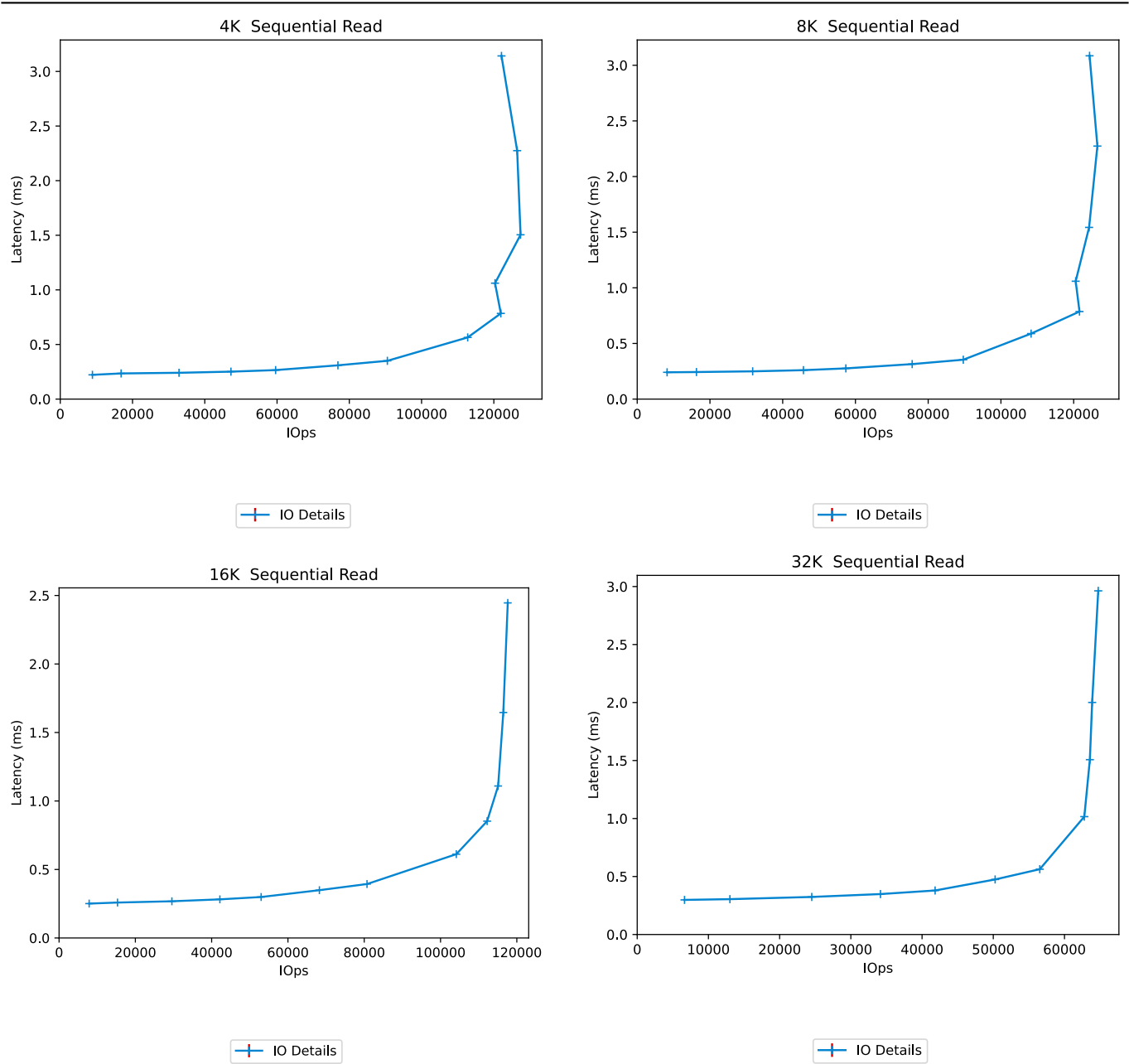
Workload Name	Number of Jobs	Maximum Throughput	Latency (ms)
4K Sequential Read	1	127395 IOps	1.5
8K Sequential Read	1	126520 IOps	2.3
16K Sequential Read	1	117623 IOps	2.4
32K Sequential Read	1	64736 IOps	3.0
64K Sequential Read	1	3573 MB/s	4.7
256K Sequential Read	1	5851 MB/s	2.9
512K Sequential Read	1	6508 MB/s	5.2
1024K Sequential Read	1	7594 MB/s	8.8
2048K Sequential Read	1	8195 MB/s	16.4
4096K Sequential Read	1	7659 MB/s	35.1
4K Sequential Write	1	477275 IOps	3.2
8K Sequential Write	1	364054 IOps	4.2
16K Sequential Write	1	209565 IOps	4.9
32K Sequential Write	1	125000 IOps	6.1
64K Sequential Write	1	4472 MB/s	7.5
256K Sequential Write	1	4990 MB/s	13.4
512K Sequential Write	1	4680 MB/s	5.3
1024K Sequential Write	1	4621 MB/s	21.5
2048K Sequential Write	1	4315 MB/s	46.0
4096K Sequential Write	1	3990 MB/s	49.1
4K Random Read	1	111420 IOps	1.1
8K Random Read	1	108488 IOps	7.1
16K Random Read	1	97223 IOps	1.3
32K Random Read	1	58928 IOps	2.2
64K Random Read	1	3626 MB/s	4.6
256K Random Read	1	6244 MB/s	5.4
512K Random Read	1	6575 MB/s	5.1
1024K Random Read	1	7803 MB/s	8.6
2048K Random Read	1	8234 MB/s	16.3
4096K Random Read	1	7771 MB/s	34.5
4K Random Write	1	24834 IOps	30.9
8K Random Write	1	23859 IOps	32.2
16K Random Write	1	22814 IOps	22.4
32K Random Write	1	21177 IOps	24.2
64K Random Write	1	1324 MB/s	19.0
256K Random Write	1	3904 MB/s	17.1
512K Random Write	1	3758 MB/s	8.8
1024K Random Write	1	3192 MB/s	7.7
2048K Random Write	1	3176 MB/s	15.4
4096K Random Write	1	3028 MB/s	43.1
4K 70/30 Random Read/Write	1	48874 IOps	5.2
16K 70/30 Random Read/Write	1	44907 IOps	5.7
64K 30/70 Random Read/Write	1	1554 MB/s	10.8

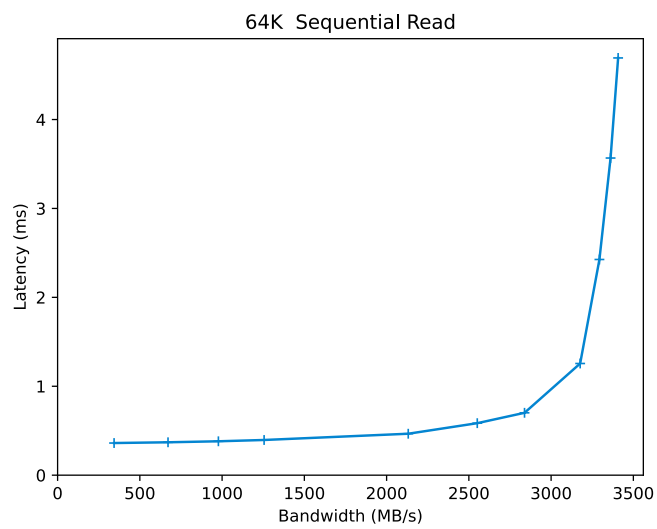
Workload Name	Number of Jobs	Maximum Throughput	Latency (ms)
64K 70/30 Random Read/Write	1	2076 MB/s	4.0

Response Curves

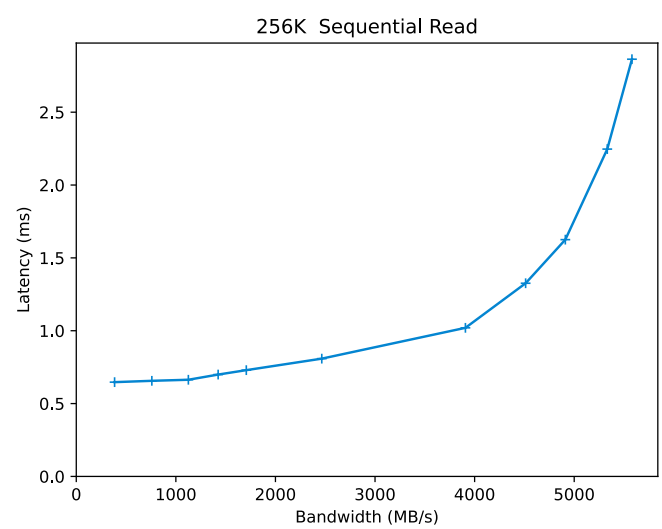
Number of Jobs 1

Sequential Read

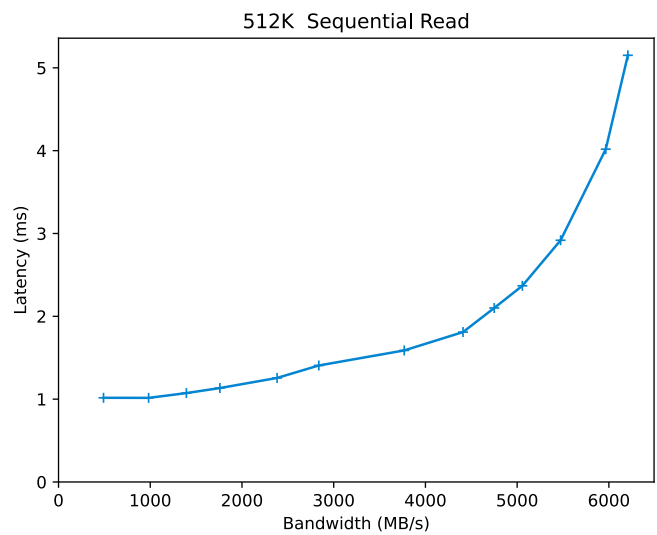




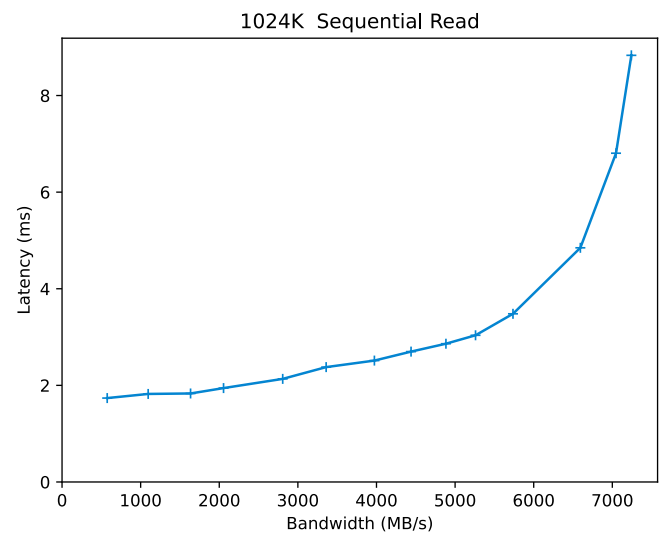
IO Details



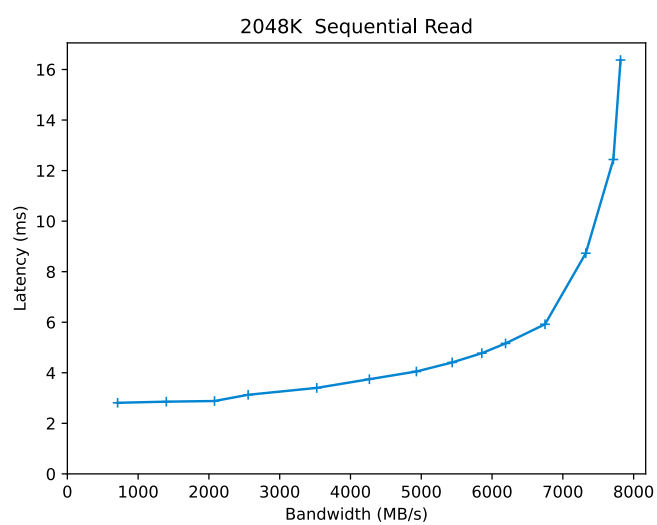
IO Details



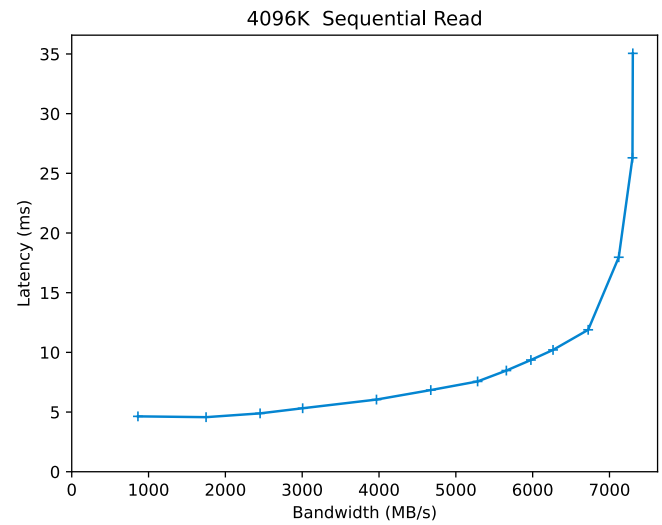
IO Details



IO Details

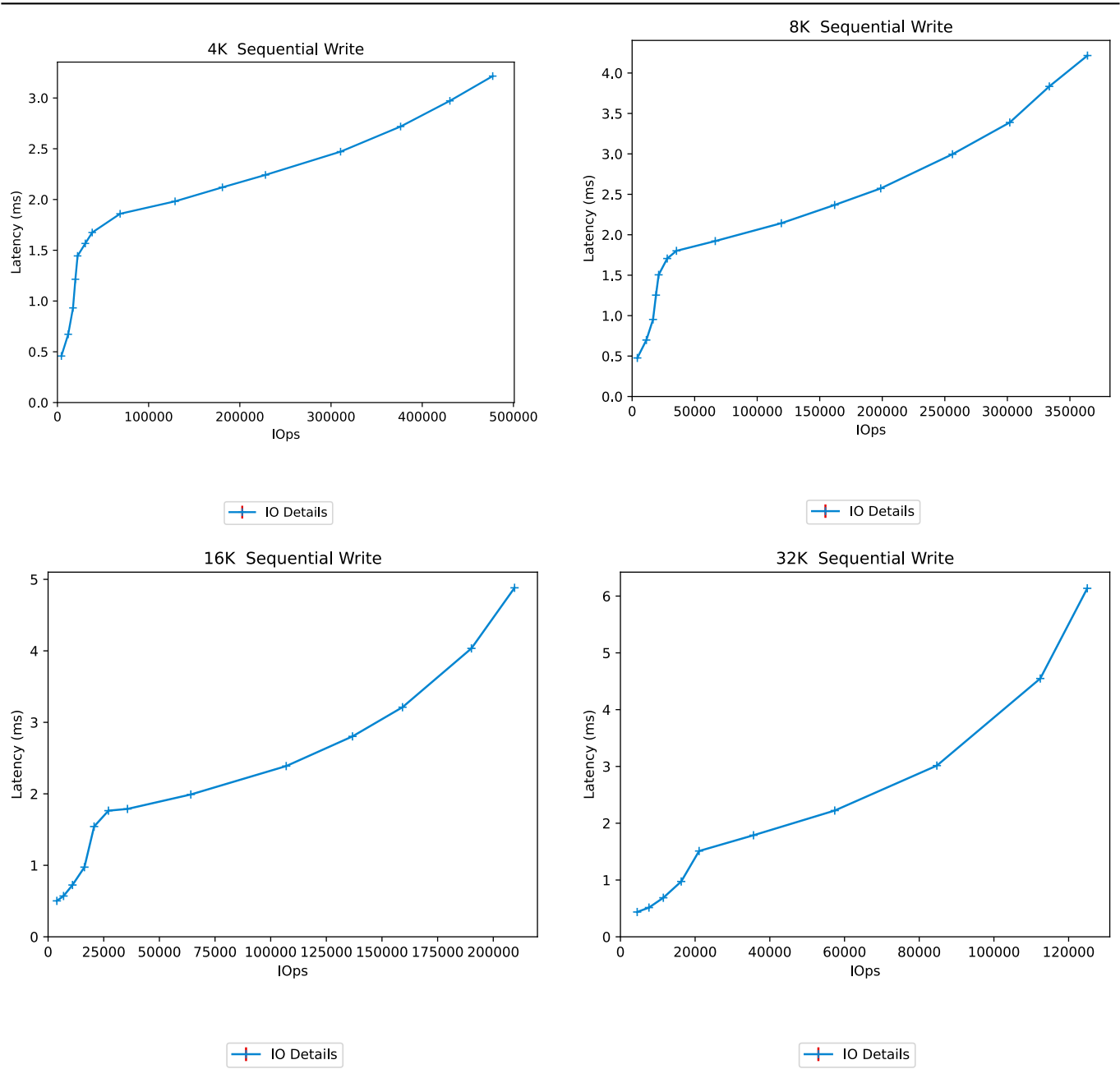


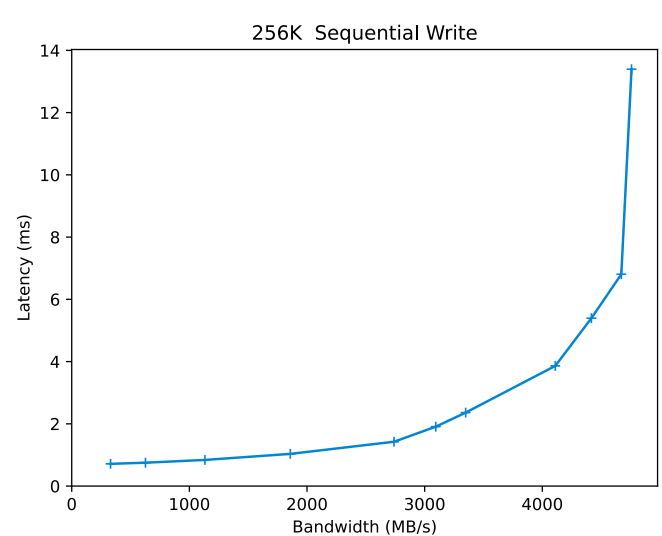
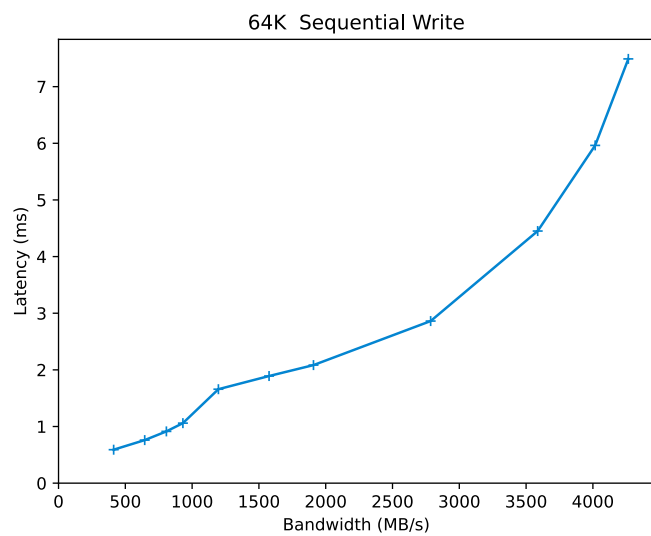
IO Details



IO Details

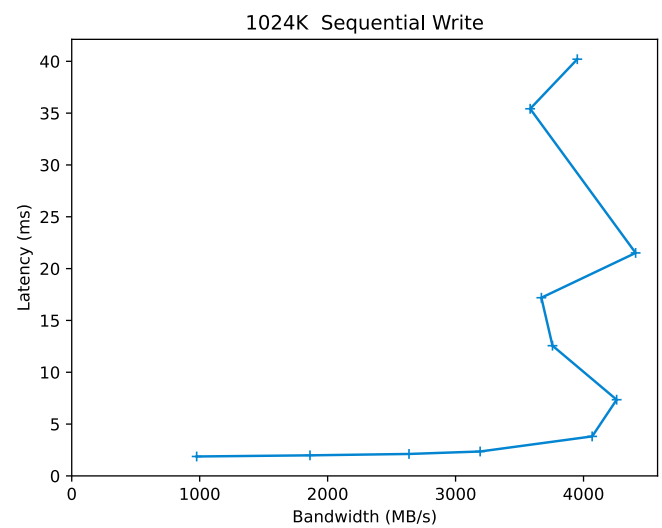
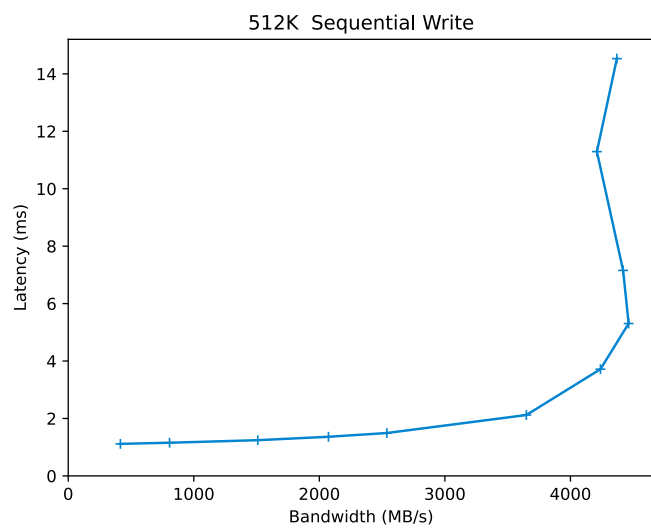
Sequential Write





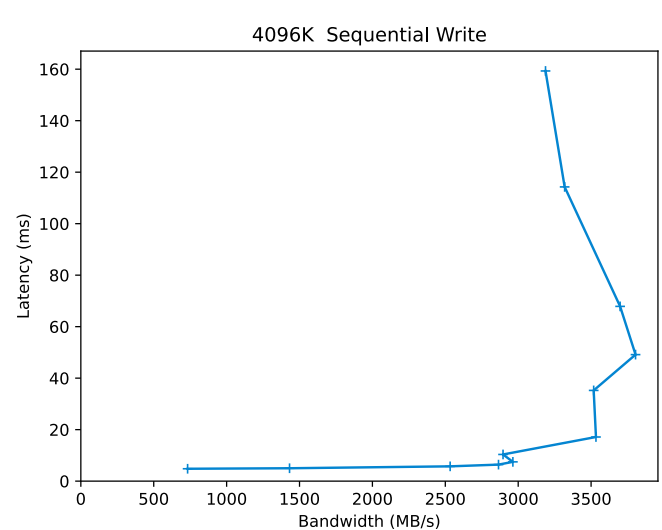
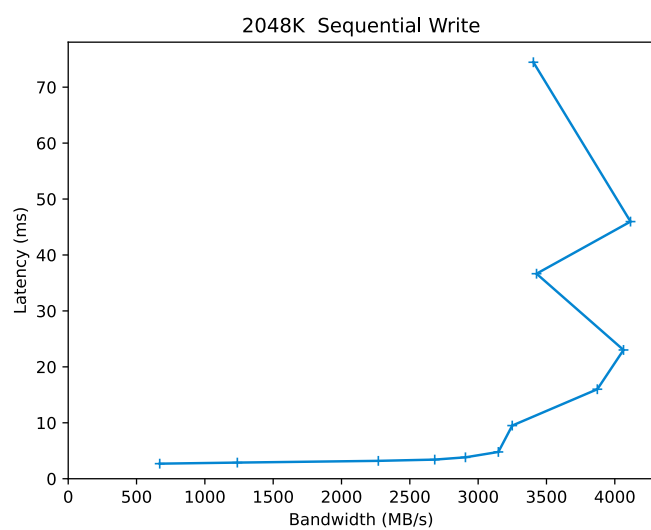
IO Details

IO Details



IO Details

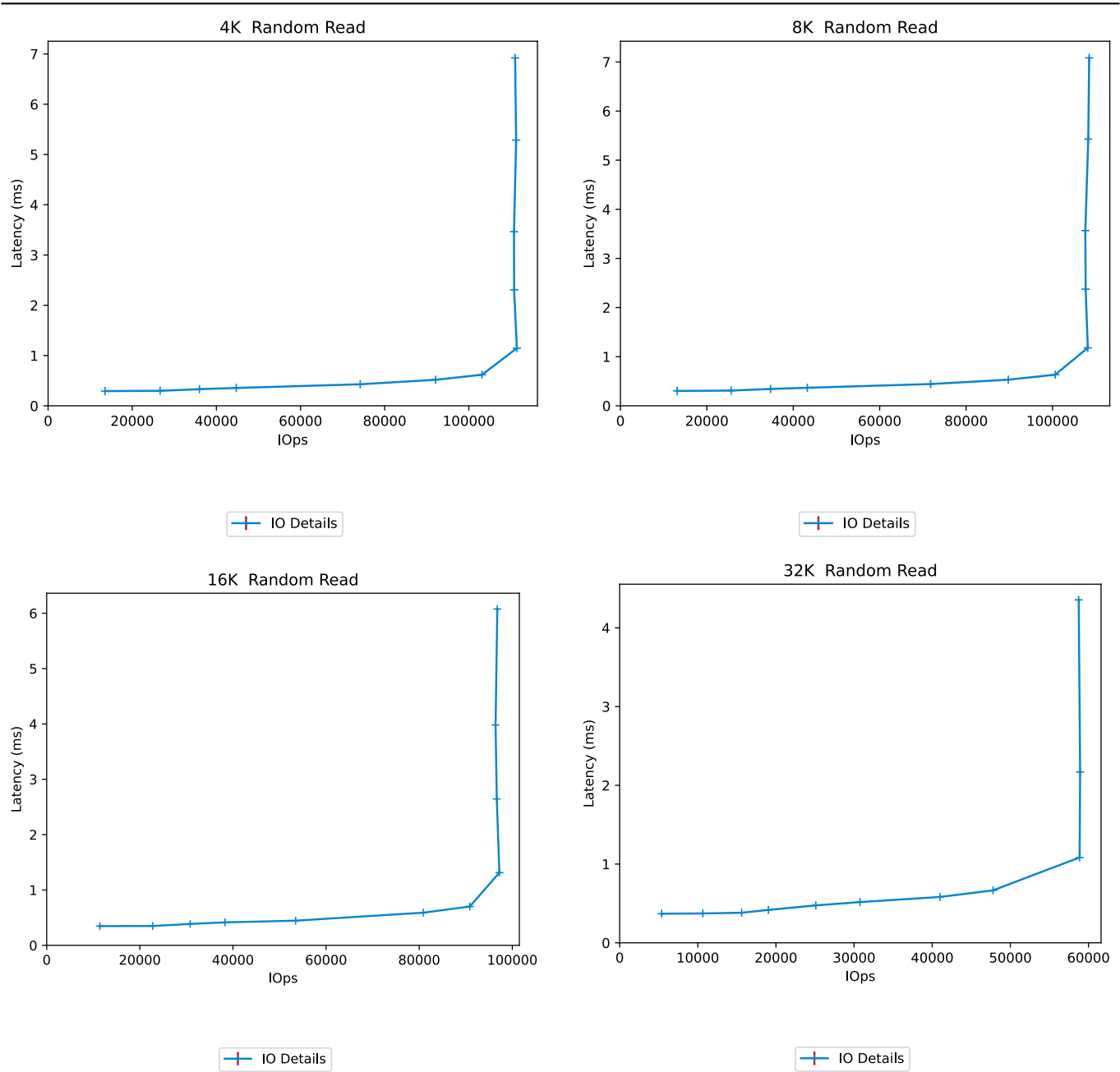
IO Details

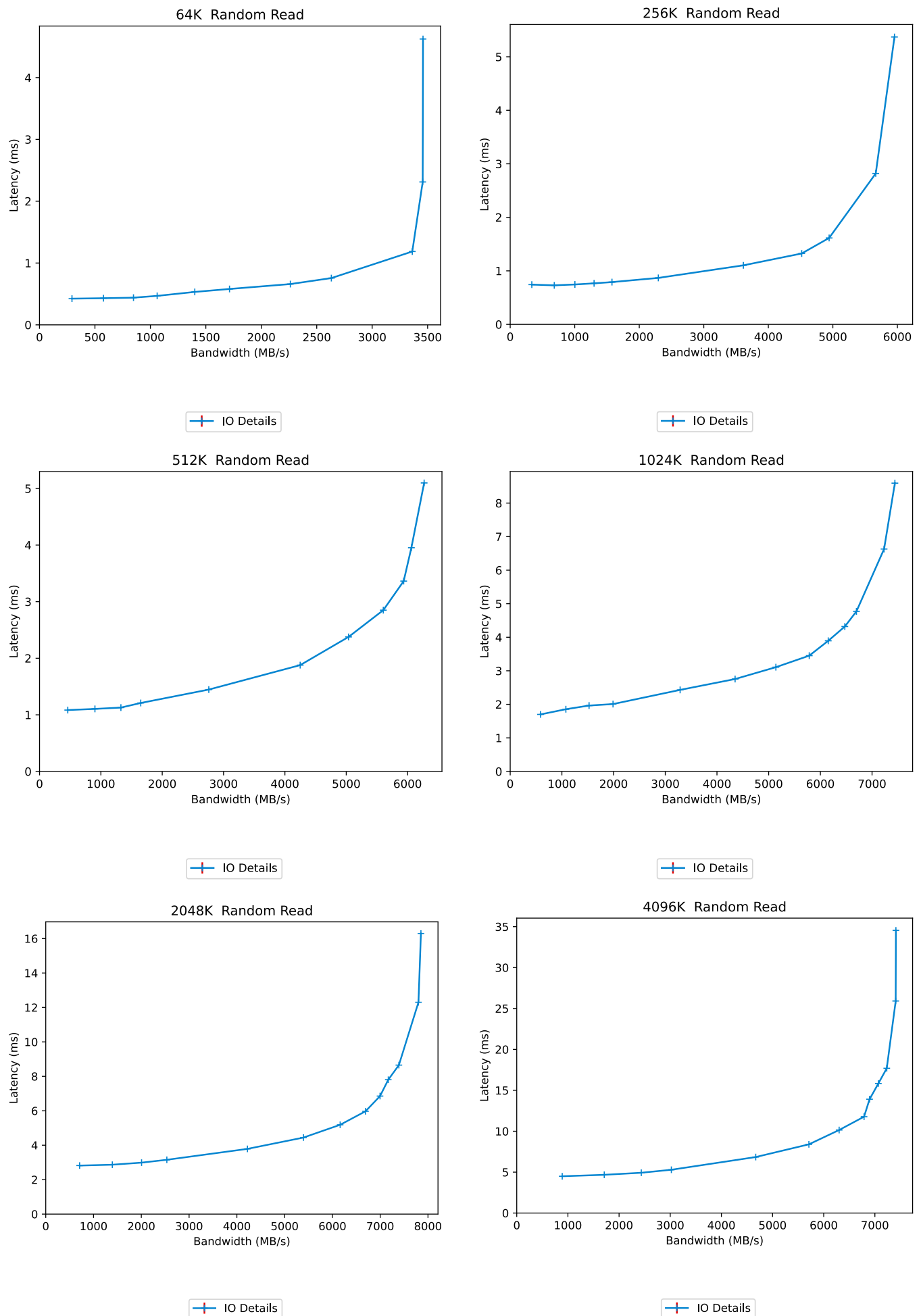


IO Details

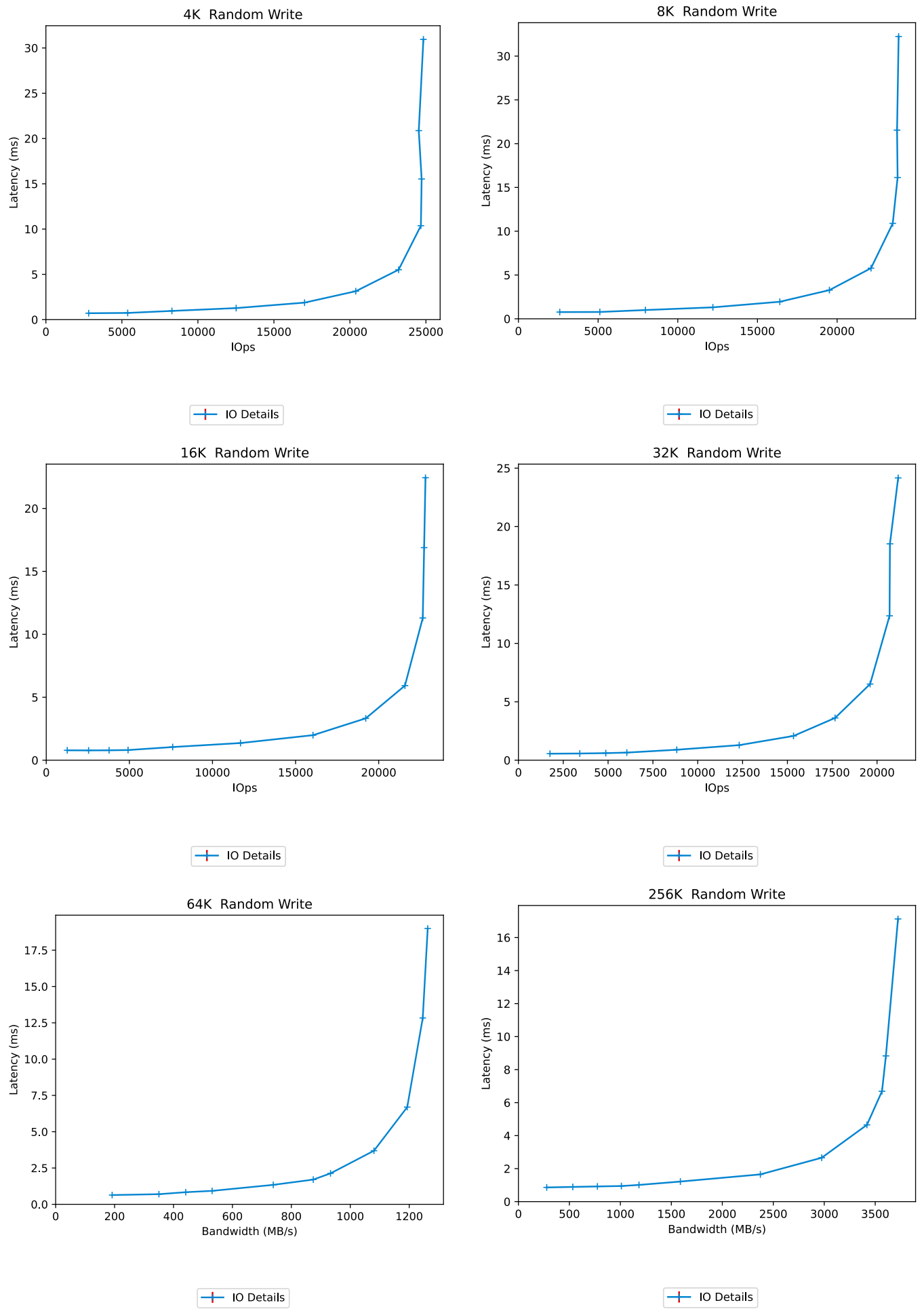
IO Details

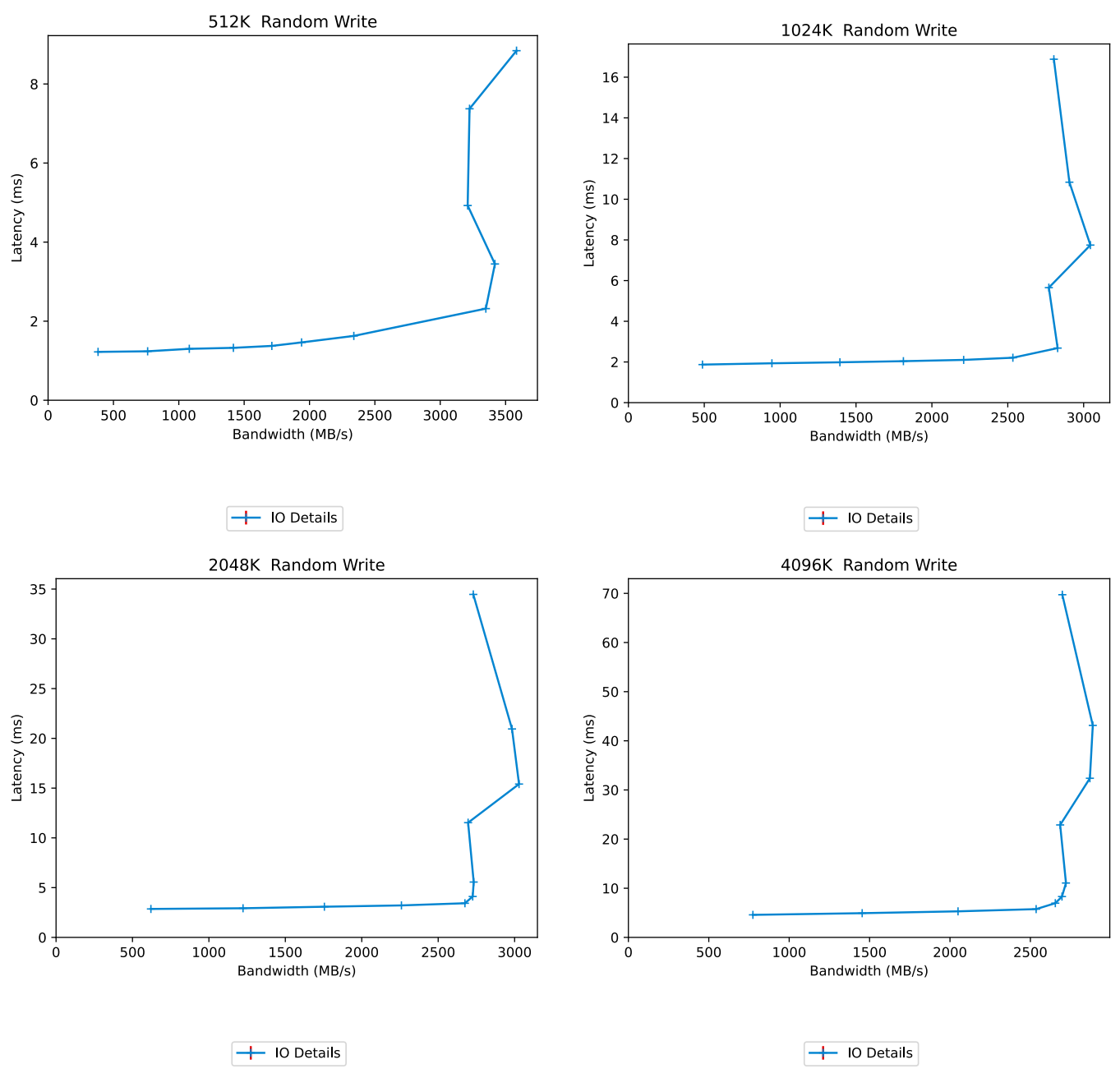
Random Read



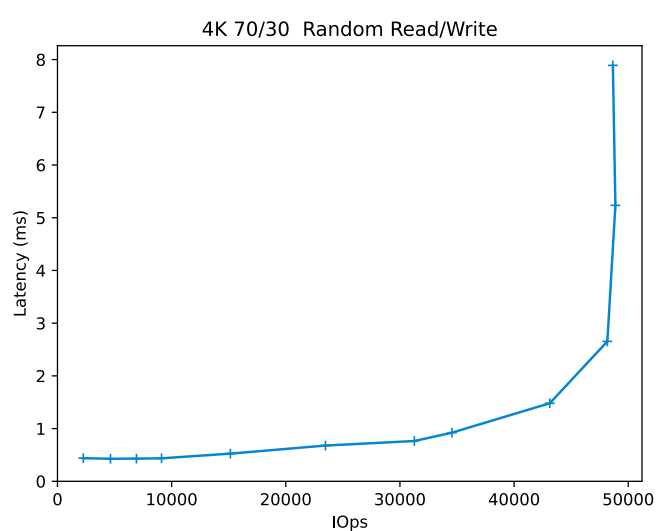


Random Write

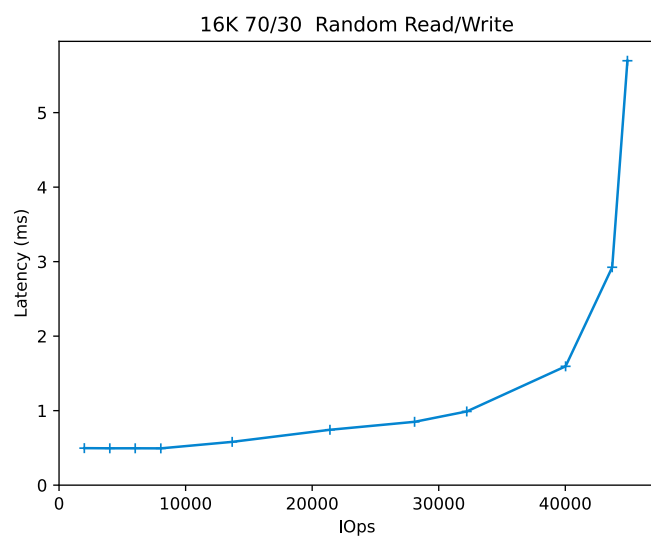




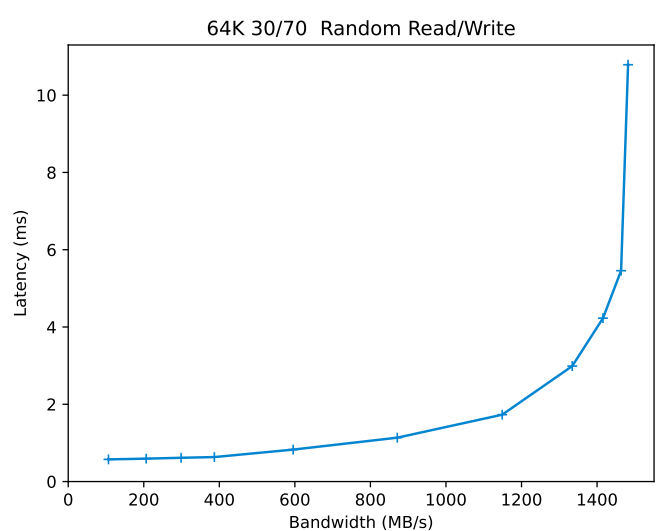
Random Read/Write



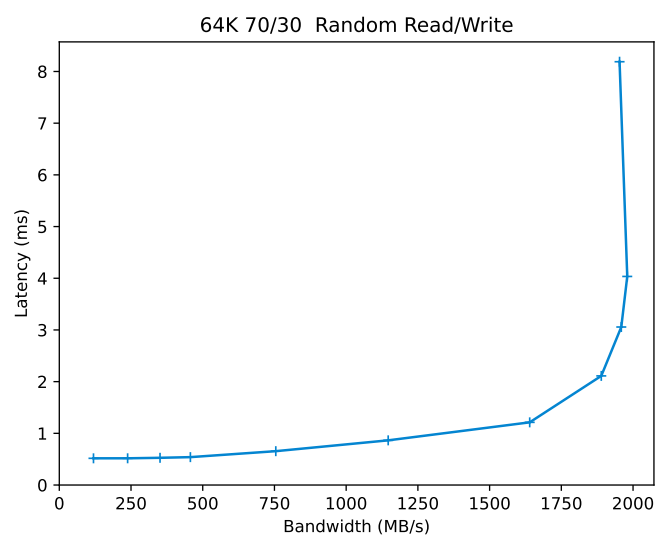
IO Details



IO Details



IO Details



IO Details

Configuration yaml

```

librbd fio:
  cmd_path: /usr/local/bin/fio.lee
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
    - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
    - 1
  ramp: 20
  rbdname: cbt-librbd fio
  time: 60
  time_based: true
  use_existing_volumes: true
  vol_size: 92812
  volumes_per_client:
    - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    16k7030:
      jobname: randmix
      mode: randrw
      numjobs:
        - 1
      op_size: 16384
      rwmixread: 70
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8
        - 16
        - 24
        - 32
        - 64
        - 128
        - 256
    16k randomread:
      jobname: randread
      mode: randread
      numjobs:
        - 1
      op_size: 16384

```

```

    total_iodepth:
      - 4
      - 8
      - 12
      - 16
      - 24
      - 48
      - 64
      - 128
      - 256
      - 384
      - 588
  16krandomwrite:
    jobname: randwrite
    mode: randwrite
    numjobs:
      - 1
    op_size: 16384
    total_iodepth:
      - 1
      - 2
      - 3
      - 4
      - 8
      - 16
      - 32
      - 64
      - 128
      - 256
      - 384
      - 512
  1Mrandomread:
    jobname: randread
    mode: randread
    numjobs:
      - 1
    op_size: 1048576
    total_iodepth:
      - 1
      - 2
      - 3
      - 4
      - 8
      - 12
      - 16
      - 20
      - 24
      - 28
      - 32
  1Mrandomwrite:
    jobname: randwrite
    mode: randwrite
    numjobs:
      - 1
    op_size: 1048576
    total_iodepth:
      - 1
      - 2
      - 3
      - 4
      - 5
      - 6
      - 8

```

```

- 16
- 24
- 32
- 48
1Mseqread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 6
    - 8
    - 10
    - 12
    - 14
    - 16
    - 20
1Mseqwrite:
  jobname: seqwrite
  mode: write
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 2
    - 4
    - 6
    - 8
    - 16
    - 32
    - 48
    - 64
    - 96
    - 128
    - 160
256krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
256krandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
    - 1

```

```
  op_size: 262144
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 32
  - 64
  - 96
  - 128
  - 256
2Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 2097152
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 12
  - 16
  - 20
  - 24
  - 28
  - 32
2Mrandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
  op_size: 2097152
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 6
  - 8
  - 16
  - 24
  - 32
  - 48
2Mseqread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 2097152
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
```

```

- 10
- 12
- 14
- 16
- 20
2Mseqwrite:
  jobname: seqwrite
  mode: write
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1
    - 2
    - 4
    - 5
    - 6
    - 8
    - 16
    - 32
    - 48
    - 64
    - 96
    - 128
32krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
32krandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 32
    - 64
    - 128
    - 256
    - 384
    - 512
4Mrandomread:
  jobname: randread

```

```

mode: randread
numjobs:
- 1
op_size: 4194304
total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
4Mrandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
  op_size: 4194304
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 6
  - 8
  - 16
  - 24
  - 32
  - 48
4Mseqread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 4194304
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
  - 10
  - 12
  - 14
  - 16
  - 20
4Mseqwrite:
  jobname: seqwrite
  mode: write
  numjobs:
  - 1
  op_size: 4194304
  total_iodepth:
  - 1
  - 2
  - 4
  - 5

```

```

- 6
- 8
- 16
- 32
- 48
- 64
- 96
- 128
4k7030:
  jobname: randmix
  mode: randrw
  numjobs:
    - 1
  op_size: 4096
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 384
4krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 4096
  total_iodepth:
    - 4
    - 8
    - 12
    - 16
    - 32
    - 48
    - 64
    - 128
    - 256
    - 384
    - 588
    - 768
4krandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
    - 1
  op_size: 4096
  total_iodepth:
    - 2
    - 4
    - 8
    - 16
    - 32
    - 64
    - 128
    - 256
    - 384

```

```

- 512
- 768
512krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
512krandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 6
    - 8
    - 16
    - 24
    - 32
    - 48
    - 64
512kseqread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 6
    - 8
    - 12
    - 16
    - 20
    - 24
    - 32
512kseqwrite:
  jobname: seqwrite
  mode: write
  numjobs:
    - 1
  op_size: 524288

```

```

    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 16
      - 32
      - 48
      - 64
      - 96
      - 128
  64k3070:
    jobname: randmix
    mode: randrw
    numjobs:
      - 1
    op_size: 65536
    rwmixread: 30
    total_iodepth:
      - 1
      - 2
      - 3
      - 4
      - 8
      - 16
      - 32
      - 64
      - 96
      - 128
      - 256
  64k7030:
    jobname: randmix
    mode: randrw
    numjobs:
      - 1
    op_size: 65536
    rwmixread: 70
    total_iodepth:
      - 1
      - 2
      - 3
      - 4
      - 8
      - 16
      - 32
      - 64
      - 96
      - 128
      - 256
  64krandomread:
    jobname: randread
    mode: randread
    numjobs:
      - 1
    op_size: 65536
    total_iodepth:
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16

```

```

- 24
- 32
- 64
- 128
- 256
64krandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
    - 1
  op_size: 65536
  total_iodepth:
    - 2
    - 4
    - 6
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 384
64kseqread:
  jobname: read
  mode: read
  numjobs:
    - 1
  op_size: 65536
  total_iodepth:
    - 2
    - 4
    - 6
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 192
    - 256
64kseqwrite:
  jobname: write
  mode: write
  numjobs:
    - 1
  op_size: 65536
  total_iodepth:
    - 4
    - 8
    - 12
    - 16
    - 32
    - 48
    - 64
    - 128
    - 256
    - 384
    - 512
8krandomread:
  jobname: randread
  mode: randread
  numjobs:

```

```

- 1
  op_size: 8192
  total_iodepth:
- 4
- 8
- 12
- 16
- 32
- 48
- 64
- 128
- 256
- 384
- 588
- 768
8krandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
- 1
  op_size: 8192
  total_iodepth:
- 2
- 4
- 8
- 16
- 32
- 64
- 128
- 256
- 384
- 512
- 768
precondition:
  jobname: precond1rw
  mode: randwrite
  monitor: false
  numjobs:
- 1
  op_size: 65536
  time: 120
  total_iodepth:
- 16
seq16kread:
  jobname: seqread
  mode: read
  numjobs:
- 1
  op_size: 16384
  total_iodepth:
- 2
- 4
- 8
- 12
- 16
- 24
- 32
- 64
- 96
- 128
- 192
- 288
seq16kwrite:

```

```

  jobname: seqwrite
  mode: write
  numjobs:
  - 1
  op_size: 16384
  total_iodepth:
  - 2
  - 4
  - 8
  - 16
  - 32
  - 48
  - 64
  - 128
  - 256
  - 384
  - 512
  - 768
  - 1024
seq256kread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 262144
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 48
  - 64
seq256kwrite:
  jobname: seqwrite
  mode: write
  numjobs:
  - 1
  op_size: 262144
  total_iodepth:
  - 1
  - 2
  - 4
  - 8
  - 16
  - 24
  - 32
  - 64
  - 96
  - 128
  - 256
seq32kread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 32768
  total_iodepth:
  - 2

```

```

- 4
- 8
- 12
- 16
- 24
- 32
- 64
- 96
- 128
- 192
seq32kwrite:
  jobname: seqwrite
  mode: write
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 2
    - 4
    - 8
    - 16
    - 32
    - 64
    - 128
    - 256
    - 512
    - 768
seq4kread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 4096
  total_iodepth:
    - 2
    - 4
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 96
    - 128
    - 192
    - 288
    - 384
seq4kwrite:
  jobname: seqwrite
  mode: write
  numjobs:
    - 1
  op_size: 4096
  total_iodepth:
    - 2
    - 8
    - 16
    - 24
    - 32
    - 48
    - 64
    - 128
    - 256

```

```

- 384
- 512
- 768
- 1024
- 1280
- 1536
seq8kread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 8192
  total_iodepth:
    - 2
    - 4
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 96
    - 128
    - 192
    - 288
    - 384
seq8kwrite:
  jobname: seqwrite
  mode: write
  numjobs:
    - 1
  op_size: 8192
  total_iodepth:
    - 2
    - 8
    - 16
    - 24
    - 32
    - 48
    - 64
    - 128
    - 256
    - 384
    - 512
    - 768
    - 1024
    - 1280
    - 1536
cluster:
  archive_dir: /tmp/cbt
  ceph-mgr_cmd: /usr/bin/ceph-mgr
  ceph-mon_cmd: /usr/bin/ceph-mon
  ceph-osd_cmd: /usr/bin/ceph-osd
  ceph-run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
    - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:

```

```

    --- server1 ---:
      a: null
mkfs_opts: -f -i size=2048
mons:
  --- server1 ---:
    a: --- IP Address ---:6789
mount_opts: -o inode64,noatime,logbsize=256k
osds:
- --- server1 ---
  osds_per_node: 8
  pdsh_cmd: /bin/pdsh
  pdsh_ssh_args: -a -x -l%u %h
  pid_dir: /tmp/cbt/pid
  rados_cmd: /usr/bin/rados
  rbd_cmd: /usr/bin/rbd
  tmp_dir: /tmp/cbt
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -oz -sZC -i 10 --sep ";" -F0 -P -f {collectl_dir}

```